

Λ -field Single Scalar Field Framework for Structure Unified Field Theory and Predictive Localized Modes in Slow-Light Systems

Author: Youfu Wang

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Abstract

We present a Λ -field single scalar field framework for the structure unified field theory, providing a predictive model for localized compression modes and boundary-locked modes in slow-light systems. The small oscillation spectrum is analyzed to assess stability, and a static one-dimensional equation is established to predict localized compression modes. A NEC-1 first-window experimental interface is proposed to facilitate future empirical validation. Gate-A1 and the second experimental window are explicitly reserved for future work, serving as a foundation for subsequent unified scaling axis construction.

Keywords: Λ -field, structure unified field theory, slow-light, localized compression mode, predictive experimental interface

1. Introduction

The structure unified field theory aims to unify local energy compression phenomena across different physical systems under a single mathematical framework. Slow-light systems provide an ideal experimental platform due to their high group refractive index and localized energy accumulation near the band edge.

In this study, we establish a single scalar Λ -field theoretical model, analyze the small oscillation spectrum, and construct a predictive framework for static localized compression modes. A first-window experimental interface (NEC-1) is defined to allow comparison with candidate experimental data. Gate-A1 and the second experimental window are explicitly reserved for future validation and for constructing a unified scaling axis.

2. Theoretical Framework

2.1 Λ -field Lagrangian

We describe the system dynamics with a single scalar field $\Lambda(x,t)$. The Lagrangian is:

$$\mathcal{L} = \frac{\tau}{2} (\partial_t \Lambda)^2 - \frac{D}{2} |\nabla \Lambda|^2 - C (\cosh(2c \Lambda) - 1) + J(x) \Lambda$$

Parameter definitions:

- τ : inertial coefficient, representing effective field inertia
- D : diffusion or coupling coefficient, characterizing spatial interaction
- C : nonlinear potential strength
- c : constant coefficient scaling the nonlinearity
- γ : dissipation coefficient in the Euler–Lagrange equation
- $J(x)$: external source term, representing localized driving or boundary perturbation

2.2 Euler–Lagrange Equation

The dynamics follow:

$$\tau \Lambda_{tt} + \gamma \Lambda_t - D \nabla^2 \Lambda + 2cC \sinh(2c\Lambda) = J(x)$$

- γ accounts for dissipation
- Boundary conditions and localized sources ($J(x)$) allow modeling of open systems, energy injection, and edge effects

3. Small Oscillation Spectrum

For small-amplitude oscillations ($\Lambda \approx 0$), the equation linearizes:

$$\tau \Lambda_{tt} - D \nabla^2 \Lambda + 4c^2 C \Lambda = 0$$

Assuming plane-wave solutions $\Lambda \sim e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, the dispersion relation is:

$$\omega^2(k) = \frac{D}{\tau} k^2 + \frac{4c^2 C}{\tau}$$

This indicates a gapped spectrum, confirming stability for small oscillations.

4. Static 1D Localized Compression Modes

The static one-dimensional equation is:

$$-D \frac{d^2 \Lambda}{dx^2} + 2cC \sinh(2c\Lambda) = J(x)$$

The Hamiltonian energy density is:

$$\mathcal{H}(x) = \frac{D}{2} \left(\frac{d\Lambda}{dx} \right)^2 + C (\cosh(2c\Lambda) - 1) - J(x)\Lambda$$

Localized compression mode criteria:

1. Peak value: $\Lambda_{\rm max} < \Lambda_c$
2. Peak width Δx located near the slow-light band edge
3. Total energy minimized: $\int \mathcal{H}(x) dx$

The theoretical curve $\Lambda_{\rm obs}(x)$ is generated for comparison with experimental data.

5. Experimental Interface (NEC-1 First Window)

- First-window reference: Ek 2014, Figure 4
- Six-point sampling method:
- Two points away from the band edge
- Two points in the transition region
- Two points near the band edge
- Experimental mapping:

$$\Lambda_{\rm obs} \sim n_g, \quad v_g = c / n_g$$

- Gate-A1 and the second experimental window are reserved for future validation.

6. Discussion

• The theoretical framework and experimental interface provide directly comparable predictions

- Localized compression modes have potential applications in:
- Photonic chip delay lines and buffers
- Phononic crystal energy localization
- Quantum light storage / EIT systems
- This version of the paper is predictive, guiding future experimental verification
- Future development, including NEC-1 second-window data and Gate-A1 cosh fitting,

will enable empirical validation and unified scaling axis construction

7. Conclusion and Future Work

- Λ -field framework establishes a complete theoretical model
- Small oscillation spectrum and localized mode analysis predict stability characteristics
- NEC-1 first-window experimental interface serves as candidate validation
- Gate-A1 and second experimental window reserved for future empirical testing

- Version v1 is suitable for Zenodo upload as an academic placeholder; version v2 can include empirical validation and updated results